HACK APERTUS 2026 · TRACK 2A

See It, Say It, Pick It

A local vision-language pipeline for safe first-pick grounding and ordered robot actions

Qwen2.5-VL + SAM 2.1 + Apertus



02

Model responsibilities

SEE

Qwen2.5-VL

objects + boxes

MASK

SAM 2.1

pixel masks

REASON

Apertus

language + plan

CHECK

Safety rules

one repair

GROUND

Geometry

interior (u, v)

Apertus decides meaning and action order. Deterministic modules enforce the physical contract.

Grounding and safety

FIRST PICK

Mask interior beats box center

Padded distance transform selects maximum clearance. Thin shapes fall back to the valid pixel nearest the mask centroid.

One object in the gripper · 85 mm opening
No trays, people, hands, unstable supports, throws or drops



04

Verification status and failure analysis

PASS

full automated suite

50

ordered public instructions

6

challenge categories

Expected failure classes

Missed small objects
Spatial relation mistakes
Planner over-refusal
Mask-edge depth errors

PUBLIC DEV SCORE

24.0 / 65
official evaluator

Held-out output: 40 / 40 IDs

05

Reproduction and submission gate

ONE COMMAND

```
powershell -File scripts/run_public.ps1 -Team YOUR_TEAM
```

01	**Pin**	Replace Apertus main with the immutable downloaded SHA
02	**Run**	Create a timestamped directory with config and Git revision
03	**Score**	Validate exact IDs, then call the official evaluator
04	**Freeze**	Update measured artifacts and run held-out once